

Corporate and Datacenter Network Architecture — Guided Notes

Unit 1.7.1 | CompTIA Network+ N10-009 Obj. 1.7 | N10-008 1.7

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

In three-tier architecture, match each layer to its primary function:

Access layer → _____

Distribution layer → _____

Core layer → _____

Question 2

A collapsed core design merges the _____ and _____ layers into one. It is appropriate for _____ environments. It becomes a problem when _____.

Question 3

In spine-leaf architecture, every _____ switch connects to every _____ switch. No _____ connects directly to another _____. Any server can reach any other server in exactly _____ hops.

Question 4

Spine-leaf exists because data centers have predominantly _____ traffic (server-to-server), while three-tier was designed for _____ traffic (client-to-server). Three-tier forces server-to-server traffic through _____ extra hops.

Question 5

In top-of-rack switching, a switch sits _____ and servers connect with _____ cables. In end-of-row switching, a larger switch serves _____ with longer cable runs. The dominant modern approach is _____ because _____.

Question 6

SDN separates the _____ plane (forwarding decisions) from the _____ plane (actual packet forwarding). A centralized _____ pushes decisions to network devices, which become simple _____.

Question 7

NFV replaces dedicated hardware _____ with software running on _____. The key advantages are _____ and _____. Examples of virtualized network functions include _____.

Question 8 — Real World Application

REAL WORLD: You're told the data center is experiencing high latency between application servers that frequently communicate. The current design is three-tier. What architecture change would you recommend and why?

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/1-7-1-corporate-and-datacenter-network-architecture/1-7-1-corporate-and-datacenter-network-architecture-answer-key.pdf
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